Project Design Phase-I Proposed Solution Template

Date	08 March 2026
Team ID	
Project Name	Ucab – Cab Booking System
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Urban commuters often face unpredictable cab availability, non-transparent fares, and limited visibility into trip status. Local drivers lack a simple digital channel to receive nearby jobs and maintain ride history. UCab addresses both sides: reliable on-demand cab booking for riders and efficient job allocation for drivers.
2.	Idea / Solution description	UCab is a MERN-based web application. Riders select pickup/drop on a Leaflet map and create a booking; drivers in the matching vehicle-type room receive the request in real-time via Socket.IO and one driver accepts. The system tracks the ride lifecycle (pending → accepted → in_progress → completed), enables rider ratings, records payments, and supports helpdesk tickets. JWT protects APIs; MongoDB stores users, drivers, bookings, payments, and support tickets.
3.	Novelty / Uniqueness	• Real-time dispatch using Socket.IO rooms (per vehicle type and per rider) to minimize fan-out and reduce noise. • Cost-effective mapping with open-source Leaflet (no proprietary map SDK lock-in). • Clean separation of concerns (Auth, Bookings, Drivers, Payments, Support) and a lightweight REST surface for fast MVP iterations. • Consistent post-ride workflows (rating, payment history, support) baked into the MVP.
4.	Social Impact / Customer Satisfaction	• Riders: reduced wait time and anxiety through live status and tracking; transparent driver ratings.

		• Drivers: improved utilization and predictable earnings via steady job inflow and ride history. • Communities: safer commutes through identity and rating trails; potential for affordable pricing with open-stack choices.
5.	Business Model (Revenue Model)	Primary: per-ride commission (e.g., 10–15% adjustable). Secondary: driver subscription for priority slots and analytics; corporate accounts for scheduled rides; promotional partnerships. (*For academic submission, these are proposed models.*)
6.	Scalability of the Solution	Stateless Express APIs with JWT support horizontal scaling; Socket.IO can use a Redis adapter for multi-instance event routing; MongoDB replica set with indexes on frequent queries (user, driver, booking status). CDN for static frontend; observability and rate limiting planned for production.